

Project Initialization and Planning Phase

Date	29 September 2026
Team ID	6ab4fddea238dc999eb665e9
Project Title	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal aims to provide an AI-powered budget planning and recommendation assistant for Home Interior, Party Planning and Jewelry needs. PocketSmart AI combines budget-aware planning with personalized Gemini-powered recommendations, helping users reduce choice overload and make suitable decisions within a fixed budget.

Project Overview	
Objective	The primary objective is to build a budget-aware GenAI assistant that helps users plan Home Interior, Party and Jewelry needs and receive personalized recommendations within their stated budget.
Scope	The project covers user authentication, budget and preference capture, three planning domains, Gemini-based recommendation generation, budget validation, recommendation details, history, and optional outfit-image analysis for jewelry recommendations.
Problem Statement	
Description	Users often face too many choices, scattered product and service information, and difficulty dividing a fixed budget across home, party or jewelry requirements. Manual comparison can make it difficult to identify suitable options quickly.
Impact	Addressing these issues can simplify planning, reduce choice overload, help users stay within their budget, and provide a more personalized recommendation experience.
Proposed Solution	
Approach	Use a FastAPI-based web application with authenticated sessions, SQLite persistence and Gemini AI to capture user requirements, generate structured recommendations, validate them against budget and preferences, and present final recommendations with details and history.
Key Features	- Home Interior, Party Planning and Jewelry planners. - Budget-aware personalized recommendations. - Gemini AI integration with structured output validation. - Optional outfit-image analysis for jewelry recommendations. - Recommendation history and detailed views. - Safe fallback when Gemini is unavailable.

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Resource Type	Description	Specification/Allocation
Hardware		
Hardware	Computing Resources	Standard laptop/PC suitable for FastAPI development and browser-based testing; GPU is not required for the core prototype because Gemini inference is provided through the API.
	Memory	8 GB RAM or higher recommended for development and local testing.
	Storage	At least 10 GB free space for source code, Python environment, SQLite database, uploads and test artifacts.
Software		
Software	Frameworks	Python, FastAPI, Uvicorn, Jinja2
	Libraries	Pydantic, SQLAlchemy/SQLite tooling, JWT/password hashing utilities, google-genai when configured, pytest
	Development Environment	VS Code or equivalent Python IDE; web browser for application testing.
Data		
Data	Data Source, size, format	User-provided budget, preferences, planning inputs and optional jewelry outfit images; stored application data is structured in SQLite. Gemini-generated recommendations are returned as structured JSON.